* Home Assignment - 1 *

A] Find first 5 partial sums for

4] $\sum_{1}^{\infty} \ln\left(\frac{n}{n+1}\right)$

$\rightarrow$

$$\ln\left(\frac{n}{n+1}\right) \Rightarrow \ln(n) - \ln(n+1)$$

$$\therefore S_K = \sum_{n=1}^{K} \left(\ln(n) - \ln(n+1)\right) = \ln(1) - \ln(K+1)$$

$$\therefore S_K = -\ln(K+1)$$

$\therefore S_1 = -\ln(2)$ $\quad \therefore S_2 = -\ln(3)$ $\quad \therefore S_3 = -\ln(4)$

$\therefore S_4 = -\ln(5)$ $\quad \therefore S_5 = -\ln(6)$

$\therefore S_1 \approx -0.6931$

$\therefore S_2 \approx -1.0986$

$\therefore S_3 \approx -1.3863$

$\therefore S_4 \approx -1.6094$

$\therefore S_5 \approx -1.7918$

$S_1 \approx -0.6931 \ldots (\ln(1/2))$

$S_2 \approx S_1 + \ln\left(\frac{2}{3}\right) \approx -\ln(3) \approx -1.0986$

$S_3 \approx S_1 + \ln\left(\frac{2}{3}\right) + \ln\left(\frac{3}{4}\right) \approx -\ln(4) \approx -1.3863$

$S_4 \approx S_1 + \ln\left(\frac{2}{3}\right) + \ln\left(\frac{3}{4}\right) + \ln\left(\frac{4}{5}\right) \approx -\ln(5) \approx -1.6094$

B] Check convergence of following by applying appropriate test (ratio or comparison):

2] $\sum_{n=1}^{\infty} \frac{e^{4n}}{(n-2)!}$

Ratio test $\Rightarrow$

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

$$\therefore a_n = \frac{e^{4n}}{(n-2)!}$$

$$\therefore \frac{a_{n+1}}{a_n} = \frac{\frac{e^{4(n+1)}}{(n-1)!}}{\frac{e^{4n}}{(n-2)!}} = \frac{e^{4(n+1)}}{(n-1)!} \cdot \frac{(n-2)!}{e^{4n}}$$

$$= \frac{e^4 (n-2)!}{(n-1)!}$$

$[\because (n-1)! = (n-1)(n-2)!]$

$$\therefore \frac{(n-2)!}{(n-1)!} = \frac{1}{n-1}$$

$$\therefore \frac{a_{n+1}}{a_n} = \frac{e^4}{n-1}$$

$$\therefore \lim_{n\to\infty} \frac{e^4}{n-1} = 0$$

$\therefore L = 0 < 1$ $\therefore$ Convergent by Ratio test c] Find interval of convergence for following power series

→ 4]

$$\sum_{n=1}^{\infty} \frac{(x+5)^n}{n(n+1)} = a_n$$

$$\therefore \frac{a_{n+1}}{a_n} = \frac{\frac{(x+5)^{n+1}}{(n+1)(n+2)}}{\frac{(x+5)^n}{n(n+1)}} = \frac{(x+5)^{n+1}}{\cancel{(n+1)}(n+2)} \cdot \frac{n\cancel{(n+1)}}{(x+5)^n}$$

$$= (x+5) \cdot \frac{n}{n+2}$$

$\therefore$ For larg 'n' $n+2 \approx n$

$$\therefore \frac{a_{n+1}}{a_n} \approx (x+5) \cdot \frac{n}{n} \Rightarrow x+5$$

Converges when, $x+5 < 1$

$\therefore -1 < x+5 < 1$

$-6 < x < -4$

$\therefore (-6, -4)$

$\therefore$ Radius of convergence is $R = 1$ & interval of convergence is $(-6, -4)$

D] Solve the following.

3] Represent $f(x) = \log(\cos x)$ as sum of its Taylor's series centered at $\pi/3$.

$f(x) = \log(\cos x)$ $\quad \therefore f(\pi/3) = \log(1/2) = -\log 2$

$f'(x) = \dfrac{1}{\cos x}(-\sin x) = -\tan x$ $\quad \therefore f'(\pi/3) = -\sqrt{3}$

$f''(x) = -\sec^2 x$ $\quad \therefore f''(\pi/3) = -4$

$f'''(x) = -2\sec^2 x \tan x$ $\quad \therefore f'''(\pi/3) = -8\sqrt{3}$

$f^{iv}(x) = -2\sec^2 x(\tan^2 x + \sec^2 x)$ $\quad \therefore f^{iv}(\pi/3) = \underline{-56}$

$$\therefore f(x) = -\log 2 - \sqrt{3}\left(x - \frac{\pi}{3}\right) - 2\left(x - \frac{\pi}{3}\right)^2 - \frac{4\sqrt{3}}{3}\left(x - \frac{\pi}{3}\right)^3 - \dots$$

$$f(x) = f\left(\frac{\pi}{3}\right) + f'\left(\frac{\pi}{3}\right)(x - \pi/3) + \frac{f''(\pi/3)}{2!}\left(x - \frac{\pi}{3}\right)^2$$

$$+ \frac{f'''(\pi/3)}{3!}(x - \pi/3)^3 + \frac{f^{iv}(\pi/3)}{4!}(x - \pi/4)^4 + \dots$$

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